

Fenrir: the first FRACTRAN cryptid, as machine 7

WS5a — the collection’s pipeline applied to a community machine four months old — July 26, 2026

1. The machine

Fenrir (March 22, 2026; Jason Yuen with Claude Opus 4.6) is three size-22 FRACTRAN programs, e.g. $[1/15, 27/77, 49/3, 10/49, 33/2]$, found in the BBf(22) cleanup and left open. The wiki gives the high-level form, and it is *literally a two-counter guarded affine machine* — our class exactly. With $S(x,y) = [x,0,0,2,y]$:

$[1,0,0,0,0] \rightarrow S(0,1) \quad | \quad S(0,2y) = \text{halt} \quad | \quad S(x,2y) \rightarrow S(x-1, 5y+2) \quad | \quad S(x,2y+1) \rightarrow S(x+2, 5y)$

Our implementation reproduces the published trajectory $S(0,1), S(2,0), S(1,2), S(0,7), S(2,15), S(4,35)$ exactly. Writing the second counter as n , the rules read

$n \text{ odd: } n \rightarrow \text{floor}(5n/2) - 2, x \rightarrow x + 2 \quad n \text{ even: } n \rightarrow \text{floor}(5n/2) + 2, x \rightarrow x - 1 \text{ (halt if } x = 0)$

so **n follows a 5/2-map and x is a walk driven by n ’s parity stream**. That is the Antihydra architecture with the multiplier 3/2 replaced by 5/2 and the parity roles exchanged: Fenrir is Antihydra’s 5/2 sibling. The wiki says only that it “follows a biased random walk”; everything below is new.

2. Reduction and theorems

theorem	statement
T1. No cycles	n is strictly increasing for every $n \geq 2$: if n is even, $5n/2 + 2 > n$; if n is odd, $(5n-5)/2 > n$ exactly when $3n > 5$. The orbit enters $n \geq 2$ at its second step, so no cycle exists and the only possible outcome besides halting is divergence. PROVED (checked to $n = 200,000$).
T2. The counting form	x is not an independent counter: $x_k = 3 \cdot O_k - k$ where O_k counts the odd n_j with $j < k$. So the whole two-counter machine collapses to the 5/2-map on n plus a counting functional of its parity stream — the analogue of the collection’s reduction to a single integer. PROVED (verified for $k < 4,000$).
T3. Exact halting criterion	Fenrir halts if and only if its orbit reaches $x = 1$ with $n \equiv 0 \pmod{4}$. Reason: $x = 0$ is reachable only from $(1, \text{even})$ by the decrement, and the new second counter $5(n/2) + 2$ is even exactly when $4 \mid n$. Equivalently, by T2: some k has $3 \cdot O_k = k + 1$ and $n_k \equiv 0 \pmod{4}$. PROVED, and brute-force verified against direct simulation on 13,500 starts (1,426 of which halt) with no mismatch.

3. Taxonomy, and the opportunity stream (P8)

Fenrir is a **coincidence-type** machine in our taxonomy: halting requires two independent conditions to coincide — a walk at a specific level and a congruence on the other counter. T3 makes the opportunity stream explicit and much sharper than a generic large-deviation estimate: **the opportunities are exactly the visits to $x = 1$** , each halting with model probability 1/4.

The walk moves +2 on odd steps and −1 on even ones, so it descends one level with probability q satisfying $q = 1/2 + q^3/2$, i.e. $q = (\sqrt{5} - 1)/2 = 0.618034$ — the golden-ratio conjugate the meta report already records for Antihydra, arising here for the same reason. From height x the chance of ever revisiting level 1 is q^{x-1} , so $\sum p_n$ converges geometrically: provbiously non-halting, on the P8 dichotomy's convergent side, with every other machine in the collection.

The verified run makes this unusually concrete. In 1,000,000 steps (n reaches 1,321,927 bits, matching $\log_2(5/2)$ per step to one part in 10^6), the odd fraction is 0.50022 and x_k/k is 0.50067 — both at their model values. The walk visits $x = 1$ **exactly once in the machine's entire history**, at step 2, in state (1, 2); it missed because 4 does not divide 2. So Fenrir has had precisely one halting opportunity, ever, and the residual risk beyond the verified prefix is $(1/4)q^{500,668}/(1 - q) \approx 10^{-104,633}$.

4. What is open, and what this adds

Open exactly where every machine in the collection is open: T3 turns halting into a single-orbit statement about the parity stream of the 5/2-map from $n = 1$, and the pseudorandomness that makes the risk estimate persuasive is an almost-everywhere fact, not a fact about this one orbit. Fenrir is a certified cryptid of the coincidence type, Π_1^0 as a single-orbit question.

Against the community's current page, this case file adds: the no-cycle theorem, the counting form, the exact halting criterion T3 (which replaces “biased random walk” with a checkable arithmetic condition), the taxonomy placement, the golden-ratio opportunity accounting, and the observation that **Fenrir and Antihydra have the identical walk and the identical constant, differing only in the digit source (5/2 versus 3/2) and in Fenrir needing an exact hit of $x = 0$ where Antihydra needs a first passage below its threshold**. That is a controlled comparison of the kind the meta report's cross-machine patterns are built from: same architecture, same risk profile, different arithmetic engine.

For the program, the transfer is the point. Fenrir was published four months ago in a different formalism (FRACTRAN) by a different community, and the collection's pipeline — reduce to a single stream, prove no cycles, extract an exact halting criterion, type it, count the opportunities — applied without modification and produced results the original analysis does not have.

5. Verification ledger

claim	status
Rules and start	From wiki.bbchallenge.org/wiki/Fenrir , fetched July 26, 2026; our implementation reproduces the published trajectory term-for-term.
T1, T2, T3	PROVED, and machine-verified: T1 to $n = 200,000$, T2 to $k = 4,000$, T3 against direct simulation on 13,500 starts.
$q = (\sqrt{5} - 1)/2$	PROVED (root in $(0,1)$ of $q^3 - 2q + 1 = 0$).
1,000,000 steps without halting; one visit to $x = 1$	MACHINE-VERIFIED (exact big-integer simulation, 247 s).
Residual risk $10^{-104,633}$	HEURISTIC. It assumes the parity stream behaves like fair coin flips; that assumption is exactly what is unproved.

claim	status
Non-halting	OPEN. Nothing here proves it.

Code: fenrir.py (rules, fidelity, T1-T2 checks), analysis.py (long run, opportunity stream, P8 accounting).